

```
%nprocshared=64
%mem=320GB
%chk=PbRhaC8C8T2.chk
# opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc
```

PbRhaC8C8T2

0 1

Pb	-0.49927595	-4.34204683	1.34599140
O	2.12775861	-2.95695229	1.70239846
O	-0.62101611	-2.00685318	0.93847349
O	2.14846695	-4.38929556	-0.59614328
C	5.65732978	-3.34440324	-0.05398687
O	5.28912124	-1.99694188	0.27385195
C	4.27818691	-1.93888614	1.25169287
C	2.92751017	-2.40754219	0.66430895
C	3.19820463	-3.47087318	-0.41391659
C	4.43532255	-4.26986814	-0.04311730
O	-0.18804663	-3.12921218	-0.91543458
C	-0.41796543	-2.04586816	-0.30595406
C	-0.44913407	-0.79021146	-1.14126099
C	-0.52146997	0.54107480	-0.41166981
O	0.55850345	0.67668111	0.54988588
C	-1.79881599	0.81349633	0.35947358
C	-3.05190347	0.70726323	-0.50708706
C	-4.21729169	1.50299379	0.06692779
C	-5.52454284	1.29217341	-0.68553360
C	-6.60592337	2.28371088	-0.27793997
C	1.78480319	0.88443538	0.07491984
O	2.06863624	0.89354699	-1.10398242
C	2.77613685	1.14479091	1.17601874
C	4.13011425	0.45806396	0.95341150
O	4.23397813	-0.67910270	1.83052586
C	6.77721802	-3.84434814	0.83806512
O	4.67908381	-5.31337681	-0.96619558
C	5.28382830	1.40747591	1.24358718
C	6.65795263	0.77710582	1.09634698
C	7.78663671	1.76509297	1.35401901
C	9.16576470	1.12519238	1.27130161
C	10.30067902	2.10484531	1.53670590
O	0.17620189	-4.18124520	4.09772902
O	-2.50420768	-3.59905355	2.44561897
O	-0.72218217	-6.57412002	3.51631269
C	-3.37284153	-5.46902174	5.89546459
O	-2.59633071	-4.38132177	6.42631064
C	-1.81805823	-3.78201270	5.41616086
C	-0.64017364	-4.69782316	5.14323413
C	-1.17018676	-6.11026071	4.78179838
C	-2.69059689	-6.12067005	4.68583947
O	-2.77369809	-4.24421921	0.35771952

C	-3.23686930	-3.76410340	1.42601729
C	-4.70129053	-3.39505453	1.51064956
C	-4.99102579	-2.00007454	2.05333472
O	-4.56058123	-1.96170358	3.43366654
C	-6.47802665	-1.69417611	1.97040457
C	-6.88217683	-0.35869880	2.57517477
C	-8.36756283	-0.08014171	2.39472359
C	-8.81165862	1.24646759	2.99341318
C	-10.28855437	1.53704651	2.76557492
C	-3.86141647	-0.90101272	3.84738082
O	-3.66235134	0.09385080	3.18794383
C	-3.26016737	-1.14227540	5.20029933
C	-1.76445862	-1.47481611	4.98893692
O	-1.34794125	-2.54226979	5.84562614
C	-4.77429678	-5.00428817	5.54429785
O	-3.18923199	-7.44216053	4.59394764
C	-0.84585943	-0.29784498	5.22941038
C	0.60122819	-0.62762944	4.89313689
C	1.56712077	0.53358403	5.08071973
C	3.02011304	0.09716440	4.94866333
C	4.01539915	1.24394774	5.04259122
H	-1.62709244	-1.81145541	3.95455244
H	-4.41846906	-1.24541402	1.50949239
H	4.20282992	0.11996432	-0.08356023
H	-0.37902777	1.32649001	-1.16105632
H	5.99937986	-3.29173357	-1.09148650
H	4.51514785	-2.60994787	2.08798287
H	2.40535582	-1.56158180	0.20457572
H	3.40593660	-2.93980798	-1.35254141
H	4.28644987	-4.69798398	0.95953372
H	0.42953281	-0.81387209	-1.78857344
H	-1.31120313	-0.87427989	-1.81053026
H	-1.86996804	0.13531437	1.21107087
H	-1.72247091	1.82869858	0.76515753
H	-2.84256643	1.07150900	-1.52200087
H	-3.34119196	-0.34507855	-0.61933994
H	-4.34861249	1.24303527	1.12255398
H	-3.95568488	2.56936395	0.04671178
H	-5.34610105	1.37850776	-1.76581254
H	-5.88147961	0.26661975	-0.52157135
H	-6.76317675	2.22553241	0.77884129
H	-6.29780397	3.27466261	-0.53866418
H	2.90919915	2.23223844	1.19335054
H	2.36204471	0.86092611	2.14294795
H	7.04905527	-4.86419204	0.55424330
H	6.47063209	-3.85647440	1.88766202
H	7.65992133	-3.20921479	0.74157391
H	3.82734322	-5.74005402	-1.12435990
H	5.18641503	2.25827468	0.55824088
H	5.16151309	1.81205488	2.25517948

H	6.74182304	-0.06876054	1.78603350
H	6.75886825	0.35101132	0.09132285
H	7.72387718	2.59245282	0.63448997
H	7.65784927	2.21710992	2.34515294
H	9.22388087	0.29775179	1.99115556
H	9.29919316	0.67168595	0.27999678
H	10.26957365	2.89151822	0.81208311
H	10.19248735	2.51881413	2.51743235
H	-3.40551893	-6.21590285	6.69408245
H	-2.41012568	-3.64920823	4.50818881
H	-0.04801857	-4.75911255	6.06332348
H	-0.86835128	-6.81840573	5.56071465
H	-2.95888475	-5.56551939	3.77790342
H	-5.14287223	-3.48415292	0.51728804
H	-5.19025945	-4.13445566	2.15539982
H	-6.75853747	-1.71952444	0.91112834
H	-7.03267586	-2.50609162	2.45741944
H	-6.63634596	-0.34712732	3.64313194
H	-6.29494316	0.44424917	2.11911252
H	-8.61086285	-0.09411456	1.32490113
H	-8.94914615	-0.89571328	2.84468554
H	-8.59766560	1.25369831	4.07029882
H	-8.21175871	2.05928370	2.56442302
H	-10.49726934	1.52205460	1.71623546
H	-10.87862207	0.79209436	3.25727239
H	-3.37240289	-0.25054196	5.81962288
H	-3.75721780	-1.97846326	5.69121849
H	-5.35348337	-5.83622648	5.13642806
H	-4.73598088	-4.20985582	4.79392931
H	-5.28756500	-4.62331981	6.42962259
H	-2.77580597	-7.84658360	3.82260024
H	-1.20186076	0.53479758	4.61309319
H	-0.93201905	0.01997690	6.27555998
H	0.93216196	-1.46116212	5.52417223
H	0.64912459	-0.97100835	3.84977580
H	1.34123613	1.32348568	4.35250703
H	1.40919532	0.98208165	6.06994118
H	3.24204715	-0.63997349	5.73187885
H	3.16912862	-0.42259331	3.99544531
H	3.83693147	1.93714434	4.24727508
H	3.89838803	1.74168022	5.98252248
H	0.50190510	-3.32850472	4.41168630
H	0.20911114	-6.81009394	3.58144223
H	1.36098145	-3.90603442	-0.91496498
H	1.96794784	-2.24749271	2.33456656
H	5.01088178	0.85957340	4.96403739
H	11.23755257	1.59266803	1.46711872
H	-7.51666676	2.04768300	-0.78758427
H	-10.52781967	2.50122629	3.16307897

C H O 0

6-311+g(2d,p)

Pb 0

S 4 1.00

591.61124370 0.22126521076D-03

46.757232559 0.56961959130D-02

20.746462696 -0.21374063831

14.610796419 0.40502620616

S 2 1.00

20.222637612 -0.83541883299D-01

6.4767324865 0.97910892388

S 1 1.00

1.6600600927 1.0000000

S 1 1.00

0.80431655001 1.0000000

S 1 1.00

0.22627039020 1.0000000

S 1 1.00

0.84014530665D-01 1.0000000

P 3 1.00

15.189102118 0.61952303583

14.693144415 -0.72498497086

6.8705890048 0.37680007984

P 3 1.00

2.2021426123 0.40196284806

1.2209125119 0.46058131862

0.63367559815 0.19367655397

P 1 1.00

0.28202837058 1.0000000

P 1 1.00

0.11333375666 1.0000000

P 1 1.00

0.43948707430D-01 1.0000000

D 6 1.00

61.315369628 0.33870800787D-03

12.372195840 0.13788683942D-01

6.9254944983 -0.75979608103D-01

2.3319539939 0.28113784298

1.2108730003 0.44474512269

0.60090478506 0.35326874351

D 1 1.00

0.28135869813 1.0000000

D 1 1.00

0.11500000000 1.0000000

F 1 1.00

1.0000000 1.0000000

F 1 1.00

0.2896200 1.0000000

PB	0	
PB-ECP	3	60
f potential		
2		
2	3.88751200	12.20989200
2	3.81196300	16.19029100
s-f potential		
4		
2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
p-f potential		
6		
2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
d-f potential		
6		
2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600
2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100